

Semester	B.E. Semester VII – CMPN
Subject	R Programming
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Laboratory	M411A

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Grade and Subject Teacher's Signature		

Experiment Number	1-NPTL
Experiment Title	To Setup and Run Introduction to R Programming, Help Functions, Packages, Libraries and RStudioTests in Jenkins Using Maven
Objectives (Skill Set / Knowledge Tested / Imparted)	To understand the basics of R programming, learn how to use help and demonstration functions, install and manage packages and libraries, and understand the use of the R command line, scripts, and RStudio.
ScreenShorts of Steps	Step 1: R Installation – Downloading R from r-project.org

Installing R

You may install R in a windows or Apple computer by downloading from <https://www.r-project.org>

Click on [download R](#)

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Step 1: R Installation – Selecting the Precompiled Binary (Windows/Mac/Linux)

Installing R

Else, download it from the Comprehensive R Archive Network (CRAN) website: <http://cran.r-project.org/>

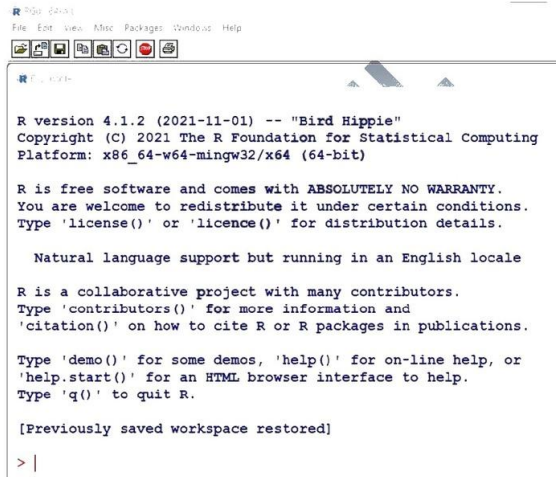
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Step 2: Starting R – RGui Console Window

Starting with R

To start R, double click on the icon .

Then we get the following Gui (Graphic user interface) window screen



```
R "Gui (64-bit)"
File Edit View Misc Packages Windows Help

R version 4.1.2 (2021-11-01) -- "Bird Hippie"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> |
```

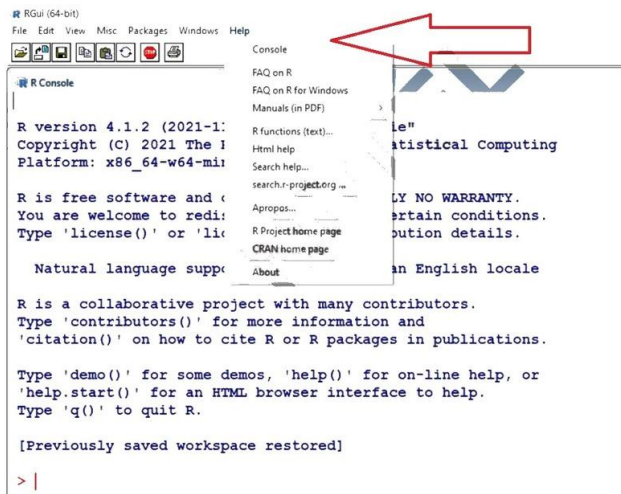
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Step 3: Accessing Help – Help Menu in RGui

Getting Help in R

This can be done in one of the following ways:

- 1) Start R software and click the help button in the toolbar of the R Gui (Graphic user interface) window.



```
R "Gui (64-bit)"
File Edit View Misc Packages Windows Help

R Console

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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> |
```

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Step 3: Accessing Help – Using ?read.table

Getting Help in R

2. Search for help in Google www.google.com

3. If you need help with a function, then type question mark followed by the name of the function. For example, **?read.table** to get help for function **read.table**.

```
R version 4.1.2 (2021-11-01) -- "Bird Hippie"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> ?read.table
starting httpd help server ... done
> |
```

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Step 3: Accessing Help – Using apropos("lm")

Getting Help in R

7) The apropos returns a character vector giving the names of all objects in the search list that match your enquiry.

apropos("lm") returns

```
> apropos("lm")
[1] ".colMeans"          ".lm.fit"            "colMeans"          "confint.lm"
[5] "contr.helmert"      "dummy.coef.lm"     "glm"               "glm.control"
[9] "glm.fit"           "KalmanForecast"    "KalmanLike"        "KalmanRun"
[13] "KalmanSmooth"      "kappa.lm"         "lm"                "lm.fit"
[17] "lm.influence"      "lm.wfit"          "model.matrix.lm"   "nlm"
[21] "nlminb"            "predict.glm"       "predict.lm"        "residuals.glm"
[25] "residuals.lm"      "summary.glm"       "summary.lm"
> |
```

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Step 3: Accessing Help – Demonstration using demo(persp)

Demonstration of R Functions

This can be useful for seeing the type of things that R can do.

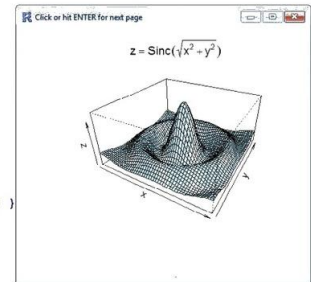
demo(persp) [**persp** is a command for 3d surface plots]

```
R >>>
> demo(persp)

demo(persp)
---- RRRRR

Type <Return> to start :

> ### Demos for persp() plots -- things not in example(persp)
> ### -----
>
> require(datasets)
> require(grDevices); require(graphics)
> ## (1) The Obligatory Mathematical surface.
> ## Rotated sinc function.
>
> x <- seq(-10, 10, length.out = 50)
> y <- x
>
> rotsinc <- function(x,y)
+ {
+   sinc <- function(x) { y <- sin(x)/x ; y[is.na(y)] <- 1; y }
+   10 * sinc( sqrt(x^2+y^2) )
+ }
>
> sinc.exp <- expression(z == Sinc(sqrt(x^2 + y^2)))
>
> z <- outer(x, y, rotsinc)
>
> oldpar <- par(bg = "white")
>
> persp(x, y, z, theta = 30, phi = 30, expand = 0.5, col = "lightblue")
Waiting to confirm page change...
```



...and it continues

Step 4: Installing & Managing Packages – packageDescription("spatial")

Description of Libraries

The command **packageDescription()** provides the description file of a package.

Here is how we find the description of the **spatial** library:

packageDescription("spatial") returns

```
R >>>
> packageDescription("spatial")
Package: spatial
Priority: recommended
Version: 7.3-14
Date: 2021-04-17
Depends: R (>= 3.0.0), graphics, stats, utils
Suggests: MASS
Authors@R: c(person("Brian", "Ripley", role = c("aut", "cre", "cph"),
+ email = "ripley@stats.ox.ac.uk"), person("Roger", "Bivand", role
+ = "ctb"), person("William", "Venables", role = "cph"))
Description: Functions for kriging and point pattern analysis.
Title: Functions for Kriging and Point Pattern Analysis
LazyLoad: yes
ByteCompile: yes
License: GPL-2 | GPL-3
```

followed by all the other details.

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Step 4: Installing & Managing Packages – install.packages("boot")


```
> install.packages("boot")
Installing package into 'C:/Users/Shalabh/Documents/R/win-library/4.1'
(as 'lib' is unspecified)
trying URL 'https://cran.curtin.edu.au/bin/windows/contrib/4.1/boot_1.3-28.zip'
Content type 'application/zip' length 641355 bytes (626 KB)

61% downloaded
URL: https://cran.curtin.edu.au/bin/windows/contrib/4.1/boot_1.3-28.zip

> install.packages("boot")
Installing package into 'C:/Users/Shalabh/Documents/R/win-library/4.1'
(as 'lib' is unspecified)
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://cran.curtin.edu.au/bin/windows/contrib/4.1/boot_1.3-28.zip'
Content type 'application/zip' length 641355 bytes (626 KB)
downloaded 626 KB

package 'boot' successfully unpacked and MD5 sums checked
The downloaded binary packages are in
C:\Users\Shalabh\AppData\Local\Temp\RtmpuAOPUm\downloaded_packages
> |
```

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Step 4: Installing & Managing Packages – installed.packages()

Which packages are installed on my computer

The command `installed.packages()` is used to see the installed packages on the computer.

```
> installed.packages()
```

```
> installed.packages()
```

Version	Package	LibPath
base	"base"	"C:/Program Files/R/R-4.1.2/library" "4.1.2"
boot	"boot"	"C:/Program Files/R/R-4.1.2/library" "1.3-28"
class	"class"	"C:/Program Files/R/R-4.1.2/library" "7.3-19"
cluster	"cluster"	"C:/Program Files/R/R-4.1.2/library" "2.1.2"
codetools	"codetools"	"C:/Program Files/R/R-4.1.2/library" "0.2-18"
compiler	"compiler"	"C:/Program Files/R/R-4.1.2/library" "4.1.2" ...

```
R Console
> installed.packages()
  Package      LibPath      Version
base         "base"      "C:/Program Files/R/R-4.1.2/library" "4.1.2"
boot         "boot"      "C:/Program Files/R/R-4.1.2/library" "1.3-28"
class        "class"     "C:/Program Files/R/R-4.1.2/library" "7.3-19"
cluster      "cluster"   "C:/Program Files/R/R-4.1.2/library" "2.1.2"
codetools    "codetools" "C:/Program Files/R/R-4.1.2/library" "0.2-18"
compiler     "compiler"  "C:/Program Files/R/R-4.1.2/library" "4.1.2"
datasets     "datasets"  "C:/Program Files/R/R-4.1.2/library" "4.1.2"
foreign      "foreign"   "C:/Program Files/R/R-4.1.2/library" "0.8-81"
graphics     "graphics"  "C:/Program Files/R/R-4.1.2/library" "4.1.2"
grDevices    "grDevices" "C:/Program Files/R/R-4.1.2/library" "4.1.2"
grid         "grid"      "C:/Program Files/R/R-4.1.2/library" "4.1.2"
KernSmooth  "KernSmooth" "C:/Program Files/R/R-4.1.2/library" "2.23-20"
```

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Step 5: Using RStudio – Downloading RStudio Desktop

Installing R Studio

https://www.rstudio.com/products/rstudio/downloads/

Getting Started Getting Started Suggested Sites Yahoo Weebill Elk dbi Citibank India Home iCloud Facebook Wikipedia

RStudio Products Solutions Customers Resources About Pricing

Download the RStudio IDE

Choose Your Version

The RStudio IDE is a set of integrated tools designed to help you be more productive with R and Python. It includes a console, syntax highlighting editor that supports direct code execution, and a variety of robust tools for plotting, viewing history, debugging and managing your workspace.

LEARN MORE ABOUT THE RSTUDIO IDE

RStudio Desktop
Desktop IDE (64-bit)
Free

RStudio Desktop Pro
Desktop IDE (64-bit)
\$995

RStudio Server
Server IDE (64-bit)
Free

RStudio Workbench
Enterprise IDE (64-bit)
\$4,975

Learn more about RStudio Team

Download and double click on the downloaded file.

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Step 5: Using RStudio – RStudio IDE Interface

Command Line versus Scripts

Use R Studio software.

RStudio

File Edit Code View Plots Session Build Debug Tools Help

Source

1

Source on Save Run Source

Environment History

Global Environment

Commands can be typed here

Defined variables are indicated here

Files Plots Packages Help Viewer

New Folder Delete Rename More

Home

- Home
- RStudio
- History
- 11111.txt
- File
- ggit
- Magi Desktop Coupon.htm
- MATLAB
- OriginLab
- R
- Parabola
- Recognition

Console

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Type 'demo()' for some demos, 'help()' for on-line help, or 'help.start()' for an HTML browser interface to help. Type 'q()' to quit R.

[workspace loaded from ~/RData]

>

This is R gui window

Conclusion	In this experiment, I learned the fundamentals of R programming and its working environment. I understood how to access help using functions such as <code>help()</code>, <code>help.search()</code>, <code>find()</code>, and <code>apropos()</code>. I also learned about R packages and libraries, including how to install, load, update, and manage them. The experiment also provided an understanding of using the R command line, scripts, and RStudio for writing and executing R programs. Thus, the basic working environment and essential tools of R programming were successfully understood.
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